Introduction to Artificial Intelligence

Lab 4: Regression and Classification

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1 Task and Dataset

The task of this lab is to build a regression model that predicts house prices using the California Housing dataset. The implementation uses `fetch_california_housing` from scikit-learn, which gives a clean tabular dataset that is suitable for a standard supervised learning experiment. The target variable is `MedHouseVal`, which represents the median house value for a district.

The dataset contains 20,640 observations and eight numerical input features: `MedInc`, `HouseAge`, `AveRooms`, `AveBedrms`, `Population`, `AveOccup`, `Latitude`, and `Longitude`. The data was split into training and testing sets using an 80/20 split. This produced 16,512 training rows and 4,128 test rows.

2 Data Preparation and Feature Analysis

The first step was to inspect the dataset and calculate summary statistics with `df.describe()`. The features are numerical, and there were no missing values that required filling or removing. The target values ranged from 0.150 to 5.000, with an average value of 2.069.

To check whether any features were clearly redundant, a correlation matrix was computed for all eight input features. The heatmap in Figure 1 shows that some features are related, especially location features and room statistics, but there is no feature that is fully dependent on another feature. For that reason, all eight features were kept for training.

Normalization was applied only where it was useful. Ridge Regression was trained inside a pipeline with `StandardScaler`, because linear models are sensitive to feature scale. Random Forest was trained without normalization, because tree-based models do not need scaled input features.

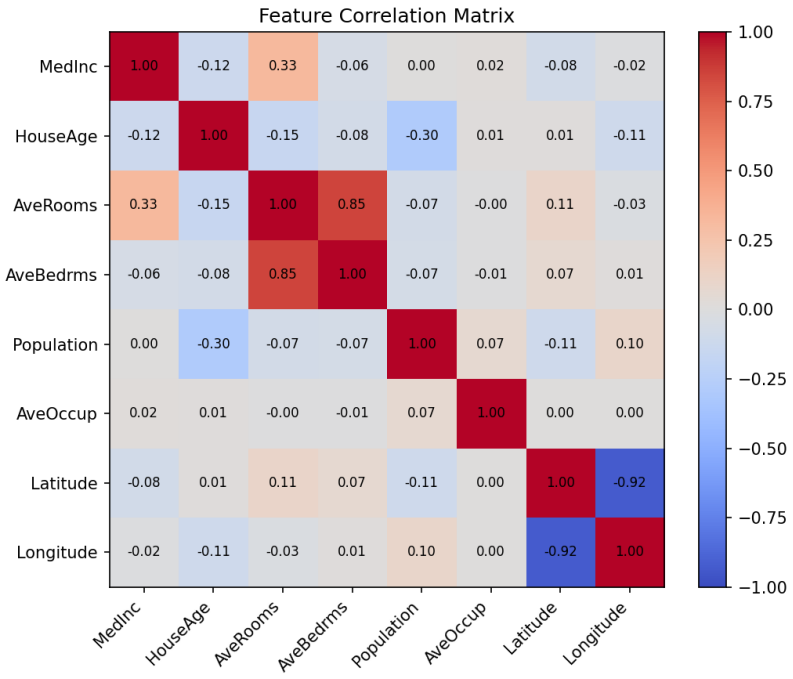


Figure 1: Correlation matrix for the eight input features.

3 Model Choice and Hyperparameters

Two regression models were selected. Ridge Regression was used as a simple linear baseline. It is easy to understand and gives a useful reference point for comparison. Random Forest Regressor was used as the second model because it can capture non-linear relationships between the input features and house prices.

The main evaluation metric was RMSE. It was used directly in `GridSearchCV` with `cv=4` and `scoring="neg_root_mean_squared_error"`. This means each hyperparameter setting was compared using 4-fold cross-validation, and the best setting was chosen by the lowest CV-RMSE.

Table 1: Ridge Regression hyperparameters tested.

`model__alpha`	CV-RMSE	CV-RMSE std
0.1	0.7214	0.0150
1.0	0.7214	0.0150
10.0	0.7214	0.0150
100.0	0.7223	0.0147

For Ridge Regression, the tested values gave very similar results. The best cross-validation result was obtained with `alpha=0.1`, although the difference from `alpha=1.0` was very small.

Table 2: Random Forest hyperparameters tested.

max_depth	min_samples_split	n_estimators	CV-RMSE
None	2	200	0.5135
20	2	200	0.5137
None	5	200	0.5139
20	5	200	0.5144
None	2	100	0.5152
None	5	100	0.5156
20	2	100	0.5157
20	5	100	0.5164
10	2	200	0.5423
10	5	200	0.5426
10	2	100	0.5440
10	5	100	0.5444

For Random Forest, the best setting used unlimited tree depth, a minimum split size of 2, and 200 trees. Smaller maximum depth, especially `max_depth=10`, clearly reduced the model's performance.

4 Results

After selecting the best hyperparameters with cross-validation, both models were evaluated on the held-out test set. The final comparison is shown in Table 3. MAE gives the average absolute prediction error, RMSE is the main metric used in this experiment, and R^2 shows how much variance in the target is explained by the model.

Table 3: Final model comparison on the test set.

Model	Test MAE	Test RMSE	Test R^2	Best 4-fold CV RMSE
Ridge Regression	0.5332	0.7456	0.5758	0.7214
Random Forest	0.3266	0.5039	0.8062	0.5135

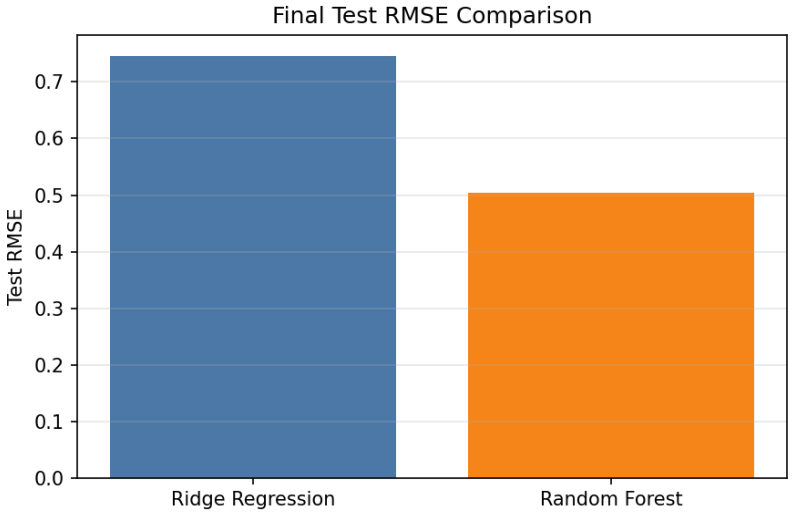


Figure 2: Test RMSE comparison for the final selected models.

The Random Forest model performed better than Ridge Regression on every reported metric. Its test RMSE was 0.5039 compared with 0.7456 for Ridge Regression, and its R^2 score was also higher. This suggests that the relationship between the features and house prices is not purely linear, so the more flexible Random Forest model is better suited to this dataset.

5 Conclusion

This lab compared a simple linear model and a non-linear ensemble model for predicting California house prices. The feature analysis did not show any fully dependent or obviously redundant variables, so all eight features were kept. Ridge Regression was useful as a clear baseline, but Random Forest achieved the best final result. Based on the lower RMSE and higher R^2, Random Forest is the preferred model for this task.